

Literature & Methodology

Albotherm's problem is not just making a capsule — it is making a capsule where the shell is flexible enough to protect the core not just at 24 hours, but to ensure quality integrity as time goes on. We saw in the Albotherm data how a capsule which was good at 24 hours will go bad as time passes. This problem was captured as drift. Hence, stability was an important issue.

We looked at the Albotherm process and EDA revealed how highly unstable the process settings were. The analysis ultimately traced that structure to three process variables: Flow Rate Ratio, Dispersed Flow Rate, and Impellor Speed. Our research dug deep into finding conditions where these processes are at optimal and can produce the best capsule. This is harder than we thought, and harder than it sounds. The relationship between these inputs and shell integrity is not completely linear. It is curved, it involves interactions between variables, and it is shaped by chemistry that does not always behave predictably. This complexity demanded us to think deep, research more, and apply greater methodological rigour.

Encapsulation chemistry and shell failure mechanisms

Before we could explore and interpret the data, we needed to understand the chemical formulations and the mechanism behind them. We asked: why would a shell fail under humidity? What determines whether a UV-cured polymer is flexible or rigid? What role does a formulation ingredient — the acrylates, the urethanes, the crosslinkers — play in how a capsule behaves?

Our understanding was shaped by the work of Lobel et al. (2024), who emphasise that the architecture of the polymer network at the molecular level determines whether a shell will flex under stress or fracture. This explains how some UV formulations in Albotherm's inputs behave. Highly crosslinked acrylate networks, such as those produced by SR833S and SR399, tend to be rigid — they fracture under humidity stress rather than flex, and that allows moisture in. This is bad for capsule integrity. On the other hand, well-performing capsules inherit urethane-acrylate systems and flexible monomers like TPGDA and ESBOA, which produce networks with more chain mobility and behave better under the same stress conditions.

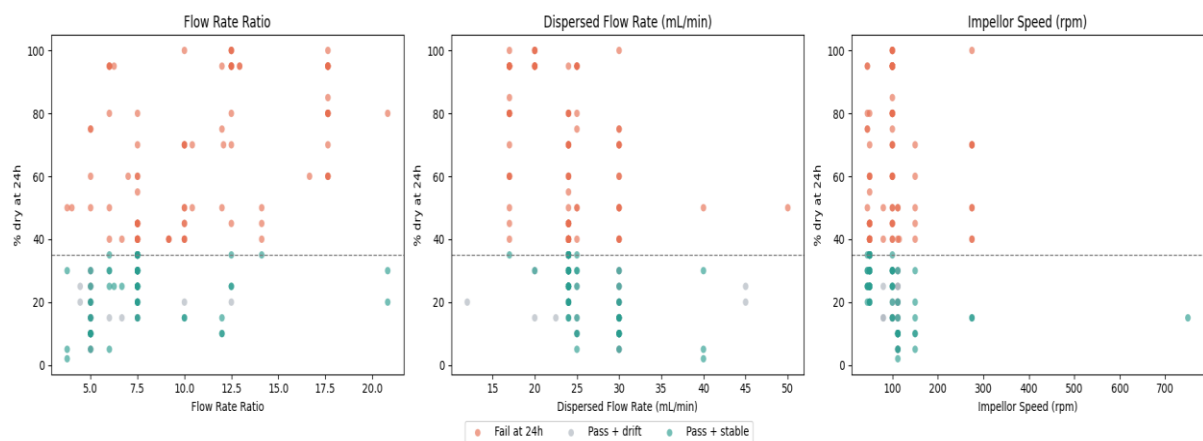
This theoretical grounding will inform future formulation decisions. To sum it all: formulation sets the ceiling, and process settings help the capsule reach it.

Why RSM — the three-step argument

Step 1: The linear model fails

We began by examining linearly the relationship between individual process variables and the durability response in isolation. The R^2 was very low at 0.035, which tells us that the distribution of durability outcomes for any given value of Flow Rate Ratio, Dispersed Flow Rate, or Impellor Speed is wide — not linear. This ruled out simple univariate relationships as the primary driver of capsule quality.

AUF110 — Process window for passing and stable capsules



OLS Regression Results

```
=====
Dep. Variable:  Q('% dry after 24h in humidity')    R-squared:                0.035
Model:                OLS                        Adj. R-squared:       0.030
Method:               Least Squares               F-statistic:           7.470
Date:                 Thu, 30 Apr 2026             Prob (F-statistic):    6.43e-05
Time:                 09:43:45                    Log-Likelihood:       -2983.2
No. Observations:     619                         AIC:                  5974.
Df Residuals:         615                         BIC:                  5992.
Df Model:              3
Covariance Type:      nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	30.0138	5.194	5.779	0.000	19.815	40.213
Q('Impellor Speed (rpm)')	-0.0004	0.012	-0.036	0.972	-0.024	0.023
Q('Dispersed Flow Rate (mL/min)')	0.2589	0.128	2.028	0.043	0.008	0.510
Q('Flow Rate Ratio')	1.3721	0.313	4.377	0.000	0.756	1.988

```
=====
Omnibus:                190.289    Durbin-Watson:           0.619
Prob(Omnibus):           0.000     Jarque-Bera (JB):         37.228
Skew:                    0.274     Prob(JB):                 8.24e-09
Kurtosis:                1.930     Cond. No.                 680.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Step 2: Interaction terms improve but are not sufficient

We then introduced pairwise interaction terms, and explanatory power increased to $R^2 = 0.173$. This gave us a clue that the three variables are not acting independently on the response. However, the model still left a substantial proportion of variance unexplained, and interaction effects alone were not sufficient to describe the relationship fully.

OLS Regression Results — Interaction Terms Model

R-squared: 0.173 Adj. R-squared: 0.164 F-statistic: 18.29 Prob(F): 3.72e-22 N: 619							
Variable	coef	std err	t	P> t	[0.025	0.975]	
Intercept	151.7275	19.805	7.661	0.000	112.834	190.621	
Flow Rate Ratio (A)	-3.4446	1.220	-2.823	0.005	-5.841	-1.048	
Dispersed Flow Rate (B)	-5.8817	0.857	-6.864	0.000	-7.564	-4.199	
Impellor Speed (C)	-1.0121	0.157	-6.431	0.000	-1.321	-0.703	
A x B (Flow Rate Ratio : Dispersed Flow Rate)	0.3478	0.070	4.989	0.000	0.211	0.485	
A x C (Flow Rate Ratio : Impellor Speed)	0.0442	0.007	6.531	0.000	0.031	0.058	
B x C (Dispersed Flow Rate : Impellor Speed)	0.0497	0.006	7.983	0.000	0.037	0.062	
A x B x C (three-way interaction)	-0.0030	0.000	-6.528	0.000	-0.004	-0.002	

Green rows = interaction terms. Dep. Variable: % dry after 24h in humidity. Covariance Type: nonrobust.

Step 3: The process window reveals curvature

The EDA extensively revealed that capsules perform better under certain process conditions, and those ranges informed a lot about what is common in good and bad production runs. Successful runs cluster within specific ranges of all three variables — failures occur predominantly at the boundaries or outside those ranges. This pattern is not a pattern of pure linear or interactive relationship. It is a curved response surface with a meaningful interior optimum — a region of maximum capsule integrity — rather than a monotone edge.

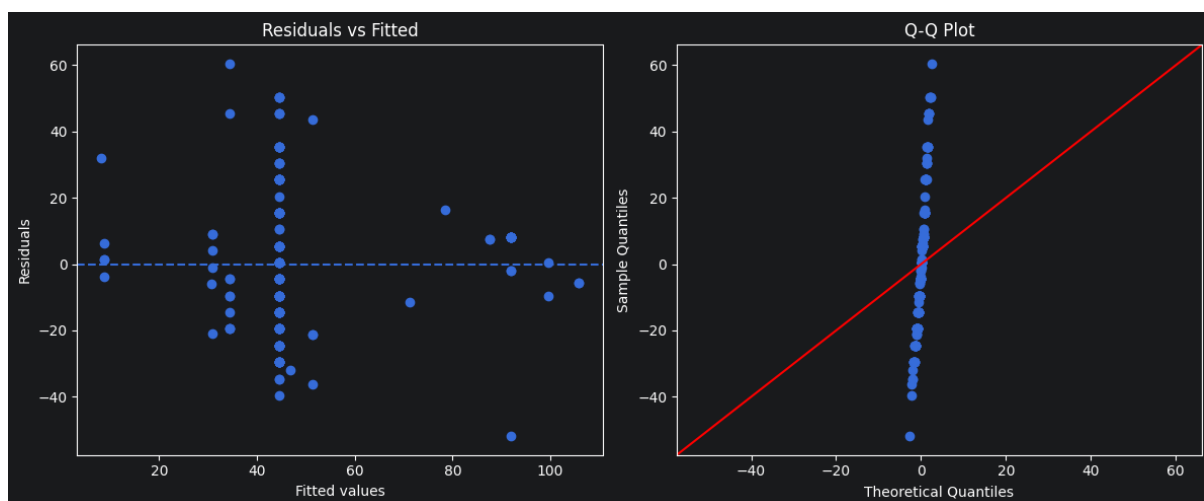
As suggested by our supervisor, we looked into RSM. Shaped by the work of Srinivasan et al. (n.d.) and Tang et al. (2010), RSM is appropriate because it handles all three relationships at the same time: linear effects, pairwise interactions, and

quadratic curvature — within a single polynomial model fitted to the historical data already available inside the process window our EDA identified.

RSM — what it found and what it cannot do

RSM is conventionally applied to purpose-designed experiments — Box-Behnken designs, Central Composite Designs — where factor levels are chosen to ensure orthogonal, balanced coverage of the design space. We applied it here to retrospective production data, which means the design points are not orthogonal and coverage of the factor space is uneven. This is reflected in the moderate R^2 values we obtained, and it is why we treat the RSM optimum as a starting point for laboratory validation rather than a final answer.

With that limitation acknowledged, RSM was still well-suited to this stage. We had 212 historical runs sitting inside the process window, and the final model — after stepwise removal of non-significant interactions — explained around 40% of the variation in 24h humidity response ($R^2 = 0.398$, adjusted $R^2 = 0.374$). The residuals showed some deviation from normality, and prediction errors for individual runs can be large.



RSM Model — OLS Regression Results							
$R^2 = 0.398$		Adj. $R^2 = 0.374$		F-statistic = 16.78		Prob(F) = 5.09e-19	
						N = 212	
Variable	coef	std err	t	P> t	[0.025	0.975]	
Intercept	-101.2795	22.979	-4.407	0.000	-146.589	-55.970	***
Flow Rate Ratio (A)	54.5810	21.413	2.549	0.012	12.360	96.802	**
Dispersed Flow Rate (B)	27.4997	8.002	3.436	0.001	11.721	43.278	***
Impellor Speed (C)	21.7378	4.754	4.573	0.000	12.364	31.111	***
A ² (Flow Rate Ratio ²)	95.2631	12.959	7.351	0.000	69.711	120.815	***
B ² (Dispersed Flow Rate ²)	110.7818	21.797	5.082	0.000	67.804	153.760	***
C ² (Impellor Speed ²)	56.0382	13.420	4.176	0.000	29.578	82.499	***
A × B (FRR : Dispersed Flow Rate)	58.3301	29.731	1.962	0.051	-0.291	116.951	~
A × C (FRR : Impellor Speed)	-12.6016	5.081	-2.480	0.014	-22.620	-2.583	**

Blue = main effects | Orange = quadratic (RSM curvature) terms | Green = significant interaction | ~ = borderline (p=0.051)
A = Flow Rate Ratio, B = Dispersed Flow Rate (mL/min), C = Impellor Speed (rpm). Dep. Variable: % dry after 24h in humidity.

But the model told us something important. The response surface is genuinely curved. A significant interaction exists between Flow Rate Ratio and Dispersed Flow Rate. And a region of minimum predicted response exists within the operating window — which is exactly what we were looking for.

RSM's deeper limitation is that it does not learn. The model is fixed once fitted and does not update as new data becomes available (Njouond Kamdem et al., 2026). In a system as complex and noisy as Albotherm's, that means the RSM optimum is best treated as an informed hypothesis to be tested in the laboratory — a direction, not a destination.

Bayesian Optimisation — where we go next

RSM told us where to look. Bayesian Optimisation is how we look more precisely.

Rather than fitting a fixed polynomial, BO builds a probabilistic surrogate model — commonly a Gaussian process — that estimates both the predicted response and the uncertainty of that prediction at any untested point in the design space (Frazier, 2018; Shahriari et al., 2016). An acquisition function then uses that uncertainty to decide where to run the next experiment, balancing exploration of unknown regions with exploitation of areas that already look promising.

What makes this different from RSM is that every new experiment makes the model smarter. The design is adaptive — it is not fixed in advance. Each run is chosen based on everything that came before it. In direct comparisons on complex process

optimisation problems, BO has shown higher predictive accuracy and lower error rates than RSM alone (Njouond Kamdem et al., 2026; Desimpel et al., 2026).

For a start-up like Albotherm, where every experimental run has a real cost, this data-efficiency is not just academically appealing. It is a commercial argument. Fewer experiments to reach the optimum means less material spent, less time lost, and faster decisions.

The plan was to use the RSM surface to initialise the Bayesian prior, then run BO iteratively to refine and validate the optimum through targeted laboratory experiments. But before BO can learn anything, it needs data that was not already in the historical record — data collected under controlled, deliberate conditions. That is what the DOE is for.

The design itself is a full 3^3 factorial — all 27 combinations of three levels across Flow Rate Ratio, Dispersed Flow Rate, and Impellor Speed — with three replicated center point runs added, bringing the total to 30 experimental runs. Center replicates serve two purposes: they allow us to detect curvature in the response surface, and they provide a pure error estimate independent of the model. The run order was randomised before being sent to the laboratory, to reduce any bias introduced by time or sequence effects. These 30 runs are what Bayesian Optimisation will learn from.

	Run	Flow Rate Ratio	Dispersed Flow Rate (mL/min)	Impellor Speed (rpm)	Run_Type
0	1	6.0	24.0	50.0	Factorial
1	2	6.0	24.0	81.0	Factorial
2	3	6.0	24.0	112.0	Factorial
3	4	6.0	27.0	50.0	Factorial
4	5	6.0	27.0	81.0	Factorial
5	6	6.0	27.0	112.0	Factorial
6	7	6.0	30.0	50.0	Factorial
7	8	6.0	30.0	81.0	Factorial
8	9	6.0	30.0	112.0	Factorial
9	10	7.0	24.0	50.0	Factorial
10	11	7.0	24.0	81.0	Factorial
11	12	7.0	24.0	112.0	Factorial
12	13	7.0	27.0	50.0	Factorial
13	14	7.0	27.0	81.0	Factorial
14	15	7.0	27.0	112.0	Factorial
15	16	7.0	30.0	50.0	Factorial
16	17	7.0	30.0	81.0	Factorial

17	18	7.0	30.0	112.0	Factorial
18	19	8.0	24.0	50.0	Factorial
19	20	8.0	24.0	81.0	Factorial
20	21	8.0	24.0	112.0	Factorial
21	22	8.0	27.0	50.0	Factorial
22	23	8.0	27.0	81.0	Factorial
23	24	8.0	27.0	112.0	Factorial
24	25	8.0	30.0	50.0	Factorial
25	26	8.0	30.0	81.0	Factorial
26	27	8.0	30.0	112.0	Factorial
27	28	7.0	27.0	112.0	Center_Replicate
28	29	7.0	27.0	112.0	Center_Replicate
29	30	7.0	27.0	112.0	Center_Replicate

One limitation we name openly: as the number of input variables grows, the search space BO must explore grows exponentially with it (Shahriari et al., 2016). Albotherm's process has many variables. Our decision was to constrain the search to the three factors the EDA identified as most influential — which keeps the dimensionality manageable and makes the BO practically implementable, not just theoretically appealing.

This is the full logic of our pipeline. EDA identified the variables. RSM modelled the surface. BO will refine the optimum. Each step earns the next one.

How the literature connects to the analysis

Paper	Where it influenced our work
Lobel et al. (2024)	Shell mechanical behaviour; mechanistic basis for the flexible vs brittle chemistry split
PMC / UV-cured encapsulation (2023)	UV-cured encapsulation stability; humidity failure in crosslinked acrylate networks
Frazier (2018)	Foundation for BO design and implementation in notebook 05
Shahriari et al. (2016)	Acquisition function design and dimensionality considerations

Njouond Kamdem et al. (2026)	Direct RSM vs BO comparison; informed the decision to sequence RSM before BO
Desimpel et al. (2026)	BO for chemical process optimisation; cost-efficiency argument for Albotherm
Tang et al. (2010)	RSM with Gaussian processes; bridging the two optimisation approaches
Srinivasan et al. (n.d.)	RSM for industrial process screening
Durante et al. (2020)	RSM applied to a real manufacturing optimisation problem

References

References

Desimpel, S., Dorbec, M., Van Geem, K.M. and Stevens, C.V. (2026) Bayesian optimization for chemical reactions. *Chemical Society Reviews*, 55, p. 2731. doi:10.1039/D5CS00962F

Durante, M. et al. (2020) Application of response surface methodology (RSM) for the optimization of supercritical CO₂ extraction of oil from Patè olive cake. *Food Chemistry*, 332, 127405.

Frazier, P.I. (2018) A tutorial on Bayesian optimization. *arXiv preprint*, arXiv:1807.02811. Available at: <https://arxiv.org/abs/1807.02811>

Lobel, B.T. et al. (2024) Current challenges in microcapsule designs and microencapsulation processes: A review. *ACS Applied Materials & Interfaces*, 16(31), pp. 40326–40355. doi:10.1021/acsami.4c07498

Njouond Kamdem, D., Kenmogne, S.B. and Wansi, J.D. (2026) Methodological approach to response surfaces and Bayesian optimization using the Gaussian process

as predictive tools for lignin extraction. *Research Square* [Preprint]. Available at: <https://doi.org/10.21203/rs.3.rs-8781383/v1>

Shahriari, B., Swersky, K., Wang, Z., Adams, R.P. and de Freitas, N. (2016) Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1), pp. 148–175. doi:10.1109/JPROC.2015.2494218

Srinivasan, K., Kumar, A., Iyer, P. and Joshi, A. (2020) Manufacturing process optimization using statistical methodologies. *arXiv preprint*, arXiv:2010.15972. Available at: <https://arxiv.org/abs/2010.15972>

Tang, Q. et al. (2010) Response surface methodology using Gaussian processes: Towards optimizing the trans-stilbene epoxidation over Co^{2+} -NaX catalysts. *Chemical Engineering Journal*, 156(2), pp. 423–431.